Scoped Boundaries for Elliptic-Curve Index Calculus: Counterexamples, Conservation, and Oracle Models

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Abstract

We assemble six proposed negative results from an existing elliptic-curve index-calculus research corpus, preserving their mathematical objects and quantifiers. A universal comparison between true and Boolean-reduced first fall degrees has explicit small counterexamples, but its interpretation depends on the degree convention and the placement of field equations. A measured failure of degree-four mutant closure does not by itself refute Semaev's F4 assumption. Table-based decomposition admits a generic simulation under explicit restrictions; connecting a concrete implementation to that model is an additional obligation. Mean decomposition multiplicity obeys an exact finite-fibre counting identity, which already has a Lean formalization in the corpus. The signed-sum description of the ordered Semaev cover gives elementary abelian monodromy and a restricted specialization law. Isogeny-map oracles are linear in the generic model; coordinate-aware oracle elimination instead requires transported challenges and an algebraic-query restriction. This is a source-bound synthesis and proof inventory. It introduces no scientific trials, claims no new theorem or novelty, and makes no deployed-curve security claim.

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1 Scope, sources, and the meaning of a negative result

An index-calculus proposal can fail for different reasons. Its counting observable can be fixed identically; its proposed generic oracle can already be simulated; an inference between degree notions can have a counterexample; or a finite solver experiment can miss its registered threshold. These are different conclusions. In particular, none establishes the nonexistence of a future coordinate-based algorithm for the elliptic-curve discrete logarithm.

The six subjects here are: Nagao's first-fall comparison; the interpretation of degree-four experiments relative to Semaev's assumption; generic simulation of compact-orbit tables; conservation of mean decomposition multiplicity; monodromy of the ordered summation cover; and isogeny-map oracles. Statements below are restatements or elementary explanations of identified existing sources. A proposition label does not assert originality or substitute for independent review. The source map and outstanding obligations are part of the draft, rather than qualifications to be removed before publication.

The corpus already contains an ECDLP programme manuscript in `crypto : papers / ecdlp-index-calculus-program/` and a conservation and specialization manuscript in `crypto-autoresearcher : papers / semaev-conservation-specialization/`. Both identify themselves as unsubmitted drafts. This companion is therefore not the first ECDLP manuscript in the programme. Its purpose is to reconcile precise scope, including contradictions between historical summaries and later source tests. It is not a publication or submission receipt.

Evidence vocabulary. A finite algebraic counterexample can disprove a universal statement, even when its variables are few. A converged closure calculation describes its specific representation and degree convention. An asymptotic generic lower bound describes an oracle model, not all computations on coordinates. An exact double count describes multiplicity, not search cost. An extrapolation of a cost model is not a measurement. We keep these distinctions throughout.

Claim chain to preserve
Mathematical object and quantifiers $\longrightarrow$ statement or frozen protocol $\longrightarrow$ proof or retained computation $\longrightarrow$ independent verification $\longrightarrow$ the conclusion with its original scope.
A missing arrow is an open obligation; a source title cannot replace it.

2 First fall comparisons and degree-four measurements

2.1 A comparison of two degree conventions

Nagao's Definitions 5 and 6 distinguish a degree in $\mathbb{F}_p[X_1, \ldots, X_N]$ from a degree after reduction modulo the field equations. Lemma 4 compares $d_F(F)$ to $d'_F(F \cup S_{\mathrm{fe}})$, with $S_{\mathrm{fe}} = \{X_i^p - X_i\}$ [2]. The frozen extraction explicitly places the field equations outside the true-side generator set in that lemma. The programme's current certificate code makes the intended comparison convention explicit [4].

For the equality convention, a fall at D has products $g_i f_i$ whose maximum degree is exactly D, and whose nonzero sum has degree strictly less than D. In the fake quantity every product and the sum is first reduced modulo S_{fe}. The requirement that the maximum is attained must be checked; a low-degree element in a span alone is insufficient.

Proposition 2.1 (Existing counterexample, equality convention)**.** *Over $\mathbb{F}_2$, the universal inequality $d_F \le d'_F$ fails. Two recorded witnesses are*

$$F_A = \{X^2Y,\ XY + X\}, \qquad F_B = \{X_1X_2 + X_3,\ X_1X_3 + X_2\}.$$

The current certificate tests assign fake degree 2 and true degree 3 to each witness, for either true-side placement of the field equations.

The witnesses and exact test assertions are in `crypto-autoresearcher : tools / lemma4 _ inside_form.py` and its test module. For F_B, take $g_1 = 1$ and $g_2 = X_1 + 1$. Boolean reduction gives

$$\overline{f_1 + (X_1 + 1)f_2} = X_2 + X_3, \quad \deg \overline{f_1} = \deg \overline{(X_1 + 1)f_2} = 2.$$

The ordinary products attain degree 3, not 2. The tests separately check all degree-two scalar combinations when field equations are adjoined, and exhibit a degree-three fall. These checks are finite certificates for the stated polynomial systems, not elliptic-curve instances or measurements of an F4 run.

The printed inequality reading is a separate axis. The source prints a $\ge$ form of the degree condition. Under the programme's padding analysis that reading collapses the quantity to one plus the least degree of a nonzero ideal element. The current test class `TestTheGeReadingVoidsOnlyOnePlacement` distinguishes the true-side placement as follows, for both witnesses:

Condition reading	True-side generators	True	Fake
Equality	F	3	2
Equality	$F \cup S_{\mathrm{fe}}$	3	2
Literal $\ge$	F	3	2
Literal $\ge$	$F \cup S_{\mathrm{fe}}$	2	2

Thus the assertion that the literal reading always voids the counterexample is too broad. The historical finding `KN-FIND-936151` contains that broader wording; the later tests explicitly correct it. Both are retained in the source inventory. The frozen extracted Lemma 4 names F on the true side, but a final publication still owes a fresh PDF-level transcription of both definitions and their intended reading. These sources do not prove or disprove Nagao's Proposition 5 by another argument, nor a complexity conclusion for ECDLP. The programme itself withholds the broader derived claim.

2.2 What the binary degree measurements actually establish

Semaev's Assumption 1 concerns d_{F4} for the Boolean system equivalent to his chained equations, with subspace dimension $k = \lceil n/m \rceil$ [12]. It is distinct from Nagao's separately numbered Assumption 1, from first fall degree, from a one-pass Macaulay rank, and from a bounded-degree mutant closure. Confusing these quantities can turn a valid computation into an invalid refutation.

The closure-at-scale study [10] reports eight unsatisfiable $m = 3$ draws at $(n, \ell) = (19, 7)$ and $(21, 7)$ for which the complete degree-four mutant closure saturates without containing 1. It also reports degree-four refutation on its other listed cells, including the $m = 4$ cells through $\ell = 4$. This is a failure of a uniform constant-four closure model in that representation. It does not follow merely from a timeout: the report says these particular closures converged. A new verifier would

still need the retained instance, closure basis, closure-stability checks and non-membership evidence before certifying an individual negative instance afresh.

Status: The broad claim that these records refute Semaev's Assumption 1 is not established.. The programme's own correction `KN-FIND-9cba50` records that an earlier ladder used Frobenius-invariant subspaces of dimension $\mathrm{ord}_n(2)$, which need not equal $\lceil n/m \rceil$, and that its degree meters were not interchangeable with d_{F4}. Conversely, the two $m = 3$ cells just listed do have $\ell = \lceil n/m \rceil$; calling them off-diagonal does not fix the metric issue. A verified bridge from saturated mutant closure to the precise F4 quantity remains necessary in either case.

There is also positive counterevidence against a naive degree-meter identification. At the separately archived $n = 17, m = 2, \ell = 9$ cell, plain degree-four Macaulay left 62 unsatisfiable targets unresolved; the subsequent mutant closure refuted all 62 with checked witnesses [6]. Therefore non-refutation by plain Macaulay was not itself evidence that mutant refutation degree exceeded four. No statement about cryptographic-size degrees follows from this finite example. Likewise, fitting either of two surviving degree-growth models does not logically refute every possible $m = 4$ degree surface outside the measured cells.

3 Compact-orbit tables: a conditional generic closure

The programme's step-back note classifies compact-orbit decomposition through enumeration, pair tables and group-law root extraction as generic [7]. The useful mathematical content is a simulation statement with explicit admissible operations.

Proposition 3.1 (Simulation schema, conditional on the interface)**.** *Suppose a prime-order-group computation constructs its advice through generic group operations and public known scalar actions. Its online phase uses only those operations, equality, and lookup against advice sets; representation-dependent indexing is required to be bookkeeping with no additional mathematical predicate. Then the table-decomposition computation can be simulated in the same generic interface, charging its advice and all simulated operations.*

The simulation tracks each available point as $[a]G+[b]Q$. Addition, negation and multiplication by a public scalar update (a, b); a table entry is another stored point, and membership against the table is an equality query or a generic lookup whose implementation cost is charged. For a fixed known factor-base point F, extraction through $S_3(x_Q, x_F, X)$ can instead compute the group elements $Q + F, Q - F$. This identity does not supply a generic method for deciding arbitrary low-degree coordinate restrictions on unknown summands. The latter operation is precisely what an algebraic decomposition method must exploit, and is outside the proposition's interface.

The generic discrete-log lower bound [13] then applies to a computation that meets these hypotheses. With target-independent advice, Corrigan-Gibbs and Kogan give an appropriate preprocessing tradeoff [1]; its frequently quoted schematic form is $ST^2 = \widetilde{\Omega}(\varepsilon r)$, with success probability ε and the original paper's units and parameter regime. An automorphism quotient needs its own model and orbit accounting before replacing r by an effective orbit count. Neither schematic inequality proves that one concrete implementation beats every other implementation in wall time or at equal bytes. Such a constant comparison needs paired measurements and an equal accounting interval.

The concrete bridge is an obligation. A factor base selected by coordinates is not automatically generated by a generic interface. A lookup key can be mere indexing, whereas a field trace, a subspace test or a coordinate-dependent branch can supply additional information. One must

inspect the frozen construction and all online predicates, state whether and how their outputs are simulated, and account for any target-independent coordinate-selected advice. This draft does not certify every current compact-pipeline path against a new simulator. In particular, the source's historical statement that all table methods are dominated by preprocessing rho at every constant must not be substituted for a same-input comparison.

The existing Lean file `GenericInvariance.lean` proves transcript invariance for a more restrictive interface: `Alg.next` sees an operation count, prior equality answers and a random tape, but no point representation. Its source comment explicitly warns that it does not cover the representation-hashing rho and kangaroo implementations that ecbench measures. It proves neither the current compact pipeline's simulation bridge nor the probabilistic preprocessing lower bound. The two model boundaries must remain visible when using that file.

4 Conservation is an exact statement about multiplicity

The conservation identity is the least assumption-sensitive result in this inventory [5]. It is a finite-fibre count.

Proposition 4.1 (Finite-pattern conservation)**.** *Let M and G be finite sets and $s : M \to G$ a function. Put $c(g) = |s^{-1}(g)|$. Then*

$$\sum_{g \in G} c(g) = |M|.$$

If G is nonempty and the target is uniform, its mean multiplicity is $|M|/|G|$.

Proof. Each element of M belongs to exactly one fibre of s. The fibres partition M, so their cardinalities sum to $|M|$. Division by the number of targets gives the stated uniform mean. □

For a finite abelian group G, a set D of B distinct elements and $m \geq 1$, choose M according to the declared counting convention:

$$\begin{aligned} \text{ordered tuples with repetition:} \quad & |M| = B^m, \\ \text{unordered multisets with repetition:} \quad & |M| = \binom{B+m-1}{m}, \\ \text{one element from each typed set } D_i : \quad & |M| = \prod_{i=1}^{m} |D_i|. \end{aligned}$$

The sum is well defined for multisets because the group is abelian. Comparing two conventions without changing the denominator is not a violation of conservation. Signed decompositions similarly require specifying whether signs are separate labels or whether $D \cup (-D)$ is a deduplicated set.

Coverage $p = |\{g : c(g) > 0\}|/|G|$ is not the mean μ. Exactly,

$$p = \mu - \frac{1}{|G|} \sum_{g} (c(g) - 1)_+ \leq \min(1, \mu).$$

When $\mu \leq 1$, equality $p = \mu$ requires every count to be zero or one; when $\mu > 1$, equality with the bound 1 requires coverage of every target and permits repeated decompositions. Conservation therefore leaves coverage, concentration and useful relation rank free. A claimed universal numerical bound on a structured base's gain over a random base would additionally require a model or bound for that random comparator; it does not follow from this double count alone.

What is not conserved. The identity is not about the cost of finding a decomposition, the rank of resulting rows, a nonuniform target law, or the mean over a selected proper subset of G. A learned filter can redistribute work across targets without contradicting it. A target-dependent base is another experiment and must charge its selection. These are genuine open axes, not exceptions to the stated identity.

The corpus already formalizes the general finite-pattern core in `FactorBaseDesigns.sum_count`. It uses a list domain and a nonduplicated target list covering its image, so domain repetitions count as labelled multiplicities. The ordinary tuple or multiset cardinality formula is a separate instantiation. Section 7 records that boundary precisely.

5 The ordered summation cover and its specialization

The relevant cover must be named before giving its monodromy. Let E/K be an elliptic curve, let $\pi : E \to \mathbb{P}^1$ be the quotient by inversion, put $n = m - 1$ with $m \geq 3$, and take independent ordered transcendentals $t_1, \ldots, t_n$. The corpus's theorem concerns

$$S_m(t_1, \ldots, t_n, T) \quad \text{over } L = K(t_1, \ldots, t_n),$$

not an unspecified symmetric parameter quotient or every specialization [3].

Proposition 5.1 (Existing signed-sum structure statement)**.** *Assume the quotient map π is separable of degree two with deck involution $[-1] \neq 1$, and the signed sums are distinct at the generic point up to simultaneous inversion. The root field and splitting field are the fixed field of diagonal inversion in $K(E^n)$, and their group over L is*

$$(\mathbb{Z}/2\mathbb{Z})^n / \langle (1, \ldots, 1) \rangle \simeq (\mathbb{Z}/2\mathbb{Z})^{m-2},$$

acting regularly on the 2^{m-2} roots.

The existing derivation obtains the product deck group from the independent quotient maps. Its roots are $x(\epsilon_1 P_1 + \cdots + \epsilon_n P_n)$. Diagonal inversion fixes all roots. A further generic coincidence would force a nonempty proper partial signed sum into $E[2]$; the preimage of that finite group scheme under the corresponding surjective sum map is proper. This gives the stabilizer exactly, and abelianness makes the single-root field the splitting field. The corpus's longer manuscript also argues that the constant field is unchanged, hence arithmetic and geometric monodromy agree. A specialist review of the quotient and characteristic-two scheme argument remains on its publication checklist; this synthesis adds no independent proof review.

Over a finite field, exclude branch points, infinite signed sums, degree drops and coincident roots. Frobenius on each degree-two fibre is a sign bit. In the regular quotient action its image has order at most two. Consequently a good specialization is either completely split or a product of equal-degree irreducible quadratics. If all inputs are abscissae of rational factor-base points, their sign bits are zero and it is completely split. This last assertion also follows directly: compute the signed sums of those rational points.

The exclusion this supplies. A factorization-type statistic is constant on this good rational factor-base locus. Such a statistic cannot distinguish relation-rate anomalies there. It does not follow that every statistic or algebraic representation is constant, or that finding the other summands for a fixed target is cheap. Complete splitting solves for the last coordinate when the others are given; relation search asks a different question.

The older finding `KN-FIND-a1f3c2` had declared a broader open problem closed. The later `KN-FIND-6edecd` explicitly declines that closure: its separate toy corroboration addresses the full-symmetric-monodromy premise, while the relation-rate and imprimitivity questions are not closed by that empirical study. We therefore report the ordered-cover structure and its stated exclusions, not the older universal closure language. Permuting or symmetrizing the input coordinates changes the base and can enlarge the monodromy; the elementary abelian description must not be transferred to that different cover without a proof.

6 Isogeny oracles: generic and coordinate-aware models

6.1 The generic statement

The source's Theorem 15.1 works with independent random encodings of order-r subgroups on curves in one rational isogeny class, where r is prime and coprime to the cofactor [8]. The action of each isogeny on chosen subgroup generators is supplied as a public scalar. A homomorphism between these cyclic subgroups is either zero or an isomorphism. Its answer on a point with formal logarithm $a + bx$ has formal logarithm $\lambda(a + bx)$ in the destination.

Thus addition, negation and map queries preserve linear formal expressions. Unless two distinct expressions collide at the uniform secret $x \in \mathbb{F}_r$, the encoded transcript supplies no equation for it. For L initial handles and Q queries, a collision union bound has the usual order $O((L + Q)^2/r)$, plus the probability of an uninformed final guess. The source writes a particular $(Q + 2)^2/r + 1/r$ expression; its exact constant requires fixing initial-handle and domain accounting. We use only the model-dependent birthday-order implication here.

This is a statement about independently encoded subgroup operations, not rational coordinate access. Known map scalars are part of its hypotheses; the cost of constructing those scalars or a useful geometric map cannot be inferred from isogeny degree alone. Pairing oracles and p-adic lifts are outside this stated interface.

6.2 Why coordinate-aware elimination needs more hypotheses

The source's later algebraic-group-model analysis [9] permits coordinate computations but requires every point submitted to an isogeny oracle to carry an algebraic representation over received points. It eliminates the oracles given a *class-joint instance*: (P_j, xP_j) with the same unknown x on every touched curve. It also gives the converse reduction. Eliminating those transported challenges further needs an explicit computable transport, for example a known spanning tree of isogenies, with its cost charged.

Accordingly, the defensible conclusion is relative equivalence to the transported challenges under the declared query restriction. Without that restriction, arbitrary coordinate-generated factor-base points can be submitted: the oracle is then a transport service for new points, and direct map evaluation has a degree-dependent cost. Without computable transport, the class-joint inputs are additional information rather than a free reduction to a single input.

The source also rejects "isogenies change constants, never exponents" as a general assertion: composite binary extension fields can contain GHS-weak and non-weak members in one isogeny class. Its prime-field exponent assertion is labelled a conjecture. Neither the generic theorem nor the coordinate-aware relative reduction resolves it.

6.3 Transported decomposition counts require the full pullback

An elementary counting result usefully separates a transfer from a new base choice. Let $\varphi : H \to H'$ be a homomorphism of finite abelian groups, let $K = \ker \varphi$, choose $D' \subseteq \operatorname{im} \varphi$ and let $D = \varphi^{-1}(D')$. For ordered decompositions and $m \geq 1$,

$$d_D(R) = |K|^{m-1} d_{D'}(\varphi R).$$

For each codomain tuple, pick arbitrary preimages of its first $m-1$ entries. The last entry is uniquely determined by requiring their sum to be R and has the required image. There are exactly $|K|^{m-1}$ choices. This is the corrected finite-group form already used in the programme manuscript [11].

The image condition and full-pullback hypothesis matter. A codomain point outside the image has no rational preimage; an arbitrarily chosen subset of a fibre need not have $|K|$ lifts. The raw transfer note's product-of-fibre-sizes display must be read tuplewise on the image, or replaced by the formula above. Counts do not conserve arithmetic cost, base membership degree, field of definition, or available endomorphisms. A statement that every representation search can only lose does not follow from this counting identity alone.

7 The two requested Lean targets

7.1 Yield conservation: a core theorem already exists

The file `crypto:research/factor_base_designs_20261015/lean/FactorBaseDesigns/Yield.lean` defines `countDphig` as the length of the filtered list of domain entries mapping to g. Its theorem `sum_count` states, exactly in mathematical notation,

$$Gs \text{ has no duplicates and contains every } \varphi(x) \text{ for } x \in D \quad \Longrightarrow \quad \sum_{g \in Gs} \operatorname{count}(D, \varphi, g) = \operatorname{length}(D).$$

It requires decidable equality of the target type, not a group law. `patterns_neutral` compares two equally long domains; `restrict_le` proves that filtering the domain cannot increase any target's multiplicity. The committed axiom log lists `propext` and `Quot.sound` for these three declarations. A committed build log reports a successful build. The present campaign also rebuilt this nine-job project with its pinned Lean 4.34.1 toolchain and reran its axiom audit; fresh logs are retained under `crypto:research/ideas_execution_20261018/runs/`.

The campaign's new `IdeasExecution.OrderedYield` module enumerates the ordered Cartesian power and proves `tuples_length`, `ordered_total_yield` and `ordered_patterns_neutral`. Its build passes; its separate axiom audit lists only `propext` and `Quot.sound`. This is an application bridge for ordered list-domain multiplicity, not a new conservation principle. For the multiset convention one still needs the enumeration length $\binom{B+m-1}{m}$ without duplicate multiset entries. Prove that the sum map and full target enumeration meet `sum_count`'s hypotheses. For a mean in $\mathbb{Q}$, require a nonempty target set before dividing. Those remaining bridges instantiate the existing theorem rather than create a new conservation principle.

7.2 Compact-pipeline genericity: an interface bridge is missing

The exact theorem `GenericInvariance.run_map` in `crypto:research/conductor_gap_cm_pairs_20261010/lean/GenericInvariance.lean` states that an injective map preserving the

declared multiplication and inversion operations commutes with every bounded run of its handle-only algorithm. `dlog_invariant` specializes it to a two-handle DLP input. The source contains no elliptic-curve construction and no probabilistic DLP lower bound. Its explicit representation-hashing limitation is binding, as discussed above. The campaign also rebuilt this four-job project and reran its axiom audit; successful checking does not extend its interface to representation hashing.

A faithful target for the compact pipeline has three layers:

1. Specify a concrete fragment with generically constructed advice, known scalar actions, membership through stored equality, and all index operations exposed. Prove an abstraction relation between concrete states and symbolic states across every executed instruction.

2. Prove a transcript simulation, including target-independent setup, lookup/multiplicity semantics, exceptional cases and the exact resource counters. Distinguish random-label hashing from the stricter handle-only theorem rather than extending that theorem by prose.

3. Import or separately formalize the appropriate generic preprocessing lower bound with its success probability, advice units, automorphism assumptions and parameter range. It is not a corollary of injective transport invariance alone.

No inspected Lean file closes all three layers for the current compact pipeline. A machine check of a small rewritten toy algorithm would not close the concrete-code correspondence. This target therefore remains open. The draft does not insert an unproved axiom or a `sorry` and call that a formalization.

8 Proof map, reproduction, and release obligations

The machine-readable `source-map.json` binds the exact repository revisions and Git objects of the principal sources. `PROOF-MAP.md` records theorem-to-source and formalization obligations in a form that can be reviewed without reading a build log. No historical result is re-scored or promoted here.

The following commands are reproduction pointers, not receipts for execution by this manuscript. Run them from the stated repository and retain new logs without overwriting prior artifacts:

- In crypto-autoresearcher, `python3-munittestdiscover-stools-ptest_lemma4_inside_form.py` checks the current certificate convention and placement table. Read the frozen primary PDF before drawing a conclusion about its intended text.

- In crypto, the closure-at-scale report binds its preregistration, raw rows and binary digests. A replay must reconstruct the same instance, check unsatisfiability independently, and verify saturated closure non-membership; a fresh generic solver timeout is not such a replay.

- In crypto's factor-base Lean project, `lakebuild`, `lakeenvleanAxiomAudit.lean`, and `lakeenvleanchecker--freshFactorBaseDesigns` are the source's reproduction commands. Preserve both the toolchain identity and fresh outputs.

- The existing conservation/specialization paper identifies its `EXP-SMON-e5cbe6` verification code and its deterministic retained run. Reproduce into a new output location before reporting new numerical verification. Finite samples remain corroboration of the derivation, not a substitute for its proof.

Before any submission or theorem promotion. Resolve the primary-text transcription and current-test disagreement for Nagao; bind a valid degree notion and instance family for any Semaev refutation; audit the compact implementation's exact generic interface; instantiate and freshly build the yield theorem; obtain a specialist check of the ordered-cover and characteristic-two arguments; and fix the exact isogeny-oracle domain/initial-handle accounting. Independently inspect every source and bibliography entry at the submission revision. Novelty, authorship, external publication and independent review are not established by this draft. Publication is a separate action with its own receipt.

The source inspection was authorized as item 19 of a twenty-item execution request. The Conductor scope check was attempted before edits and returned `conductor:commandnotfound`. The session had no knowledge-retrieval tool and used the indexed primary repository artifacts instead. These are disclosed tooling limits, not findings about elliptic curves. Drafting this manuscript partially advances item 19; it does not complete its publication and formalization gates.

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